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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: The study was approved by the Animal Ethics Committee of the University of Western Australia and performed in accordance with Principles of Laboratory Care (National Health and Medical Research Council Australian code for the care and use of animals for scientific purposes , eighth edition , 2013) .
Euthanasia Agents: The mice were subsequently euthanized by cervical dislocation , and tissues were collected.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Male age- and littermate-matched wild-type (wt) and homozygous mutant mice (N434S) were housed in standard cages (45 cm by 29 cm by 12 cm) under a 12-hour lightdark schedule (lights on 7:00 a . m . to 7:00 p . m .) in controlled environmental conditions of 22deg + 2degC and 50 + 10 relative humidity .
Subject Demographics
Age: Metabolic studies Six-week-old male mice were fed either a NCD or HFD for 14 weeks .
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication

Type: The data were results from three independent biological experiments , and all significantly enriched proteins are shown in black and red with 0.01 false discovery rate
Number: The data were results from three independent biological experiments , and all significantly enriched proteins are shown in black and red with 0.01 false discovery rate
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Immunoblotting Specific proteins were detected using the following rabbit monoclonal antibodies: phospho-SAPKJNK (Thr183Tyr185 , 4668) , SAPKJNK (9252) ,	phospho-SAPKJNK		
phospho-Akt (Ser473 , 4060) , Akt (9272) , S6 ribosomal protein (2217S) , phospho-S6 ribosomal protein (4856S) , glyceraldehyde phosphate dehydrogenase (GAPDH) (2118) , LC3AB (12741) MCU (14997) (Cell Signaling Technologies) , and MRPP3 (HPA020459) (Sigma-Prestige) diluted 1:1000 rabbit polyclonal antibodies: GLUT2 (54460) (Abcam) diluted 1:1000 , GLUT4 (PA-1-1065) (Thermo Fisher Scientific) , CPTII (OAAN00972) (Aviva System Biology) diluted 1:500	phospho-Akt (Ser473 , 4060) , Akt (9272		
	phospho-S6 ribosomal protein		
	GAPDH		
	MRPP3 (HPA020459		
	GLUT2		
	GLUT4		
	PA-1-1065		
	OAAN00972		
, LRPPRC (sc66844) (Santa Cruz Biotechnology) diluted 1:500 , TFAM (ab131607) (Abcam) diluted 1:1000 mouse monoclonal antibodies: total OXPHOS Cocktail Antibody (ab110412) , VDAC (ab14734) , and SDHA (ab14715) (Abcam) , diluted 1:1000	LRPPRC		
	VDAC		
	SDHA		
IRDye 800CW goat anti-rabbit immunoglobulin G (IgG) or IRDye 680LT goat anti-mouse IgG (LI-COR Biosciences) secondary antibodies were used , and the immunoblots were visualized using an Odyssey infrared imaging system (LI-COR Biosciences) .	anti-rabbit immunoglobulin G (IgG		
	anti-mouse IgG		
Experimental Models: Cell Lines			
Bimolecular fluorescence complementation NIH-3T3 cells were transfected with separate plasmids encoding LETM1 with the VN fragment (amino acids 1 to 173 of Venus) attached to the C-terminal end and wt or N434S MRPP3 with the VC fragment (amino acids 174 to 239 of Venus) attached to the C-terminal end and mCherry red	NIH-3T3		Suggestion: RRID:CVCL_0594(link)

fluorescent protein (RFP) in a 2:2:1 ratio .			
Recombinant DNA			
BioID MRPP3 and its N434S mutant were cloned into pcDNA3.1-MCS-BirA(R1186)-HA .	pcDNA3.1-MCS-BirA(R1186)-HA		
Software and Algorithms			
fruit fly , Drosophila melanogaster , NP_001245549.1 mosquito , Anopheles gambiae , XP_003435980.1 frog , Xenopus tropicalis , XP_002935379.1) , and the alignment was produced using Clustal Omega .	Clustal Omega		Suggestion: Clustal Omega (RRID:SCR_001591)(link)
The local electrostatic potential was modeled using the Adaptive Poisson-Boltzmann Solver program and visualized using PyMOL (Schrodinger) .	Solver		Suggestion: Solver (RRID:SCR_008510)(link)
	PyMOL		Suggestion: PyMOL, RRID:SCR_000305(link)
The area under or above the curve was calculated using the trapezoidal rule using Excel software (Microsoft) .	Excel		
Biologically independent reads (three from each genotype and treatment) were aligned against the mouse genome GRCm38.p6 primary assembly (ftp:ftp.ebi.ac.ukpubdatabasesgencodeGencode_mouserelease_M21gencode.vM21 . primary_assembly . annotation . gtf . gz ftp:ftp.ebi.ac.ukpubdatabasesgencodeGencode_mouserelease_M21gencode.vM21 . transcripts . fa . gz) masked for nuclear mitochondrial sequences with STAR v2.7.0f (56) and the GENCODE vM23 gene annotation with a customized mitochondrial annotation .	GENCODE		
Full fragment length mitochondrial coverage profiles were generated with samtools v1.10 (57) and bedtools v2.26.0 (58) and normalized to total reads mapped to the mouse genome.	samtools		
	bedtools		
Sequenced reads were trimmed via TrimGalore v0.6.5 using cutadapt (54) 0.18 and FastQC (55) v0.11.8 in paired-end mode with a minimum length of 18 nt .	TrimGalore		
	FastQC		
Biologically independent reads (three from each genotype and treatment) paired reads were merged using NGmerge v0.3 (59) .	NGmerge		
Reads were aligned to the GRCm38.p6 genome sequence	STAR		

using STAR (56) and the GENCODE vM21 gene annotation (with customized mitochondrial annotations) .			
Fluorescent images were taken at 37degC before and after addition of 10 mM glucose using an Andor Zyla SCMOS 5.5 MP camera attached to an inverted Nikon TE2000-U microscope , driven by Metamorph 7.10 software (excitation , 545 nm emission , 620 nm interval , 0.5 min exposure , 50 ms) .	Metamorph		Suggestion: MetaMorph Microscopy Automation and Image Analysis Software (RRID:SCR_002368)(link)

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq